A DERIVATION EXTENDS ALONG A FORMALLY SMOOTH LOCAL HOMOMORPHISM OF COMPLETE NOETHERIAN LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Ring of dual numbers over a ring and derivations as maps into it. used at proof lines 12010-12016. Supplies the target of the lifting problem and the identification of the epsilon component with a derivation.
- Formal smoothness in the adic topology through the lifting of diagrams along square-zero ideals. used at proof lines 12017-12020. Provides the map from the formally smooth algebra into the dual numbers.
- Continuity criterion for a lift along a formally smooth local map. the author cites Lemma lemma-lift-continuous at proof line 12017. Ensures that the lifted map is a map of topological rings, which is what the formal smoothness hypothesis requires.

Source reference. Original lemma 07PL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9801–9801. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07NJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9764–9764. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a local homomorphism of Noetherian local rings. Let $D : A \to A$ be a derivation. Assume that B is complete and $A \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology. Then there exists an extension $D' : B \to B$ of D.*

Proof. Denote $B[\epsilon] = B[x]/(x^2)$ the ring of dual numbers over B. Consider the ring map $\psi : A \to B[\epsilon]$, $a \mapsto a + \epsilon D(a)$. Consider the commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{1} & B \\ \uparrow & & \uparrow \\ A & \xrightarrow{\psi} & B[\epsilon] \end{array}$$

By Lemma 07NJ and the assumption of formal smoothness of B/A we find a map $\varphi : B \to B[\epsilon]$ fitting into the diagram. Write $\varphi(b) = b + \epsilon D'(b)$. Then $D' : B \to B$ is the desired extension. □